

Active Information Gathering in Social Deduction Game

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Abstract

Social deduction games provide a controlled testbed for studying agent reasoning and action selection under uncertainty and potential deception. These games often require belief inference to reason about hidden information. Existing work treats belief inference largely as a passive diagnostic: beliefs are updated as evidence arrives, but the maintained belief state is not used to actively plan what to do next. In this work, we study active information gathering for reducing uncertainty in the belief state in Tofu Kingdom, a social deduction game structured around a sequence of questions, constrained answers, and a single guess. We decompose the questioning agent into three modules: possible-worlds belief tracking, question selection, and final guessing; and implement multiple methods for each. The methods are combined into two agents: a basic symbolic agent using Bayesian updates over possible worlds, information-gain driven question selection, and MAP guessing; and an LLM agent that performs all three sub-tasks through structured prompting. We report preliminary experimental results showing that (1) possible-worlds belief tracking improves accuracy regardless of question strategy; (2) LLM-based question selection outperforms the information-gain heuristic only when given access to belief context; and (3) an opponent-model mismatch is the dominant failure mode of the symbolic agent against strategic opponents. These findings motivate a future neuro-symbolic design in which each sub-task is delegated to the component best suited for it.

Introduction

In real-world interactions, people often need to infer others’ beliefs and intentions from noisy observations that may be incomplete, misleading, or deliberately deceptive. Social deduction games provide a controlled testbed for studying this challenge of reasoning and decision-making under uncertainty and deception. In games such as Avalon and Tofu Kingdom (Ultraboardgames 2026; Avalon-game.com 2026), players are assigned hidden roles with different factional objectives, and some may lie to achieve those objectives. The well-defined rules and discrete structure of these games enable controlled experiments while preserving the core difficulty of strategic reasoning in deceptive multi-agent settings.

Some early work used symbolic or neuro-symbolic methods for social deduction games. Eger and Martens (Eger and Martens 2017, 2018) built a Dynamic Epistemic Logic

reasoning system for One Night Ultimate Werewolf, representing game states as Kripke models with epistemic updates over possible worlds; and DeepRole (Serrino et al. 2019) combines counterfactual regret minimization with deep value networks to play Avalon. These systems, however, rely on structured actions rather than natural-language communication.

Recently, large language models (LLMs) have been widely adopted as autonomous agents across a broad range of applications, including conversational assistance, code generation, and decision support. This progress has also motivated researchers to develop LLM-based agents for social deduction games. One research direction uses LLMs as game-playing agents, but prompting alone remains insufficient for strategic deception: AvalonBench (Light et al. 2023) shows a GPT-3.5 agent underperforming a simple rule-based bot, and ReCon (Wang et al. 2023) adds structured prompting while leaving reasoning internal to the LLM with no explicit consistency check. More broadly, LLMs still struggle with structured logical reasoning (Liu et al. 2025), especially in settings involving incomplete information and deception (Wang et al. 2023). This motivates a hybrid direction that combines LLMs with symbolic methods for belief inference, such as GRAIL (Rahimirad et al. 2025) and CSP4SDG (Xu et al. 2025), as part of a broader effort to use symbolic structure to compensate for LLM limitations (Yang et al. 2025).

Despite this progress, existing approaches treat belief inference mainly as a passive diagnostic tool: they update beliefs as new evidence arrives, but do not use those beliefs to guide information-gathering actions. Consequently, it remains unclear both how belief states can support active information gathering in social deduction games and whether such actions improve performance on broader game objectives.

Within the broader trend of using symbolic methods to enhance LLM reasoning and planning (Yang et al. 2025), this project therefore focuses on improving the information-gathering behaviour of LLM-based agents in social deduction games. Two questions guide our investigation: how an agent can actively gather information in a social deduction game where other agents may deceive, and whether information-gathering actions can improve agent performance?

To address these questions, we make three contributions. First, we decompose the main agent (the Prince) into three modules: belief tracking, question selection, and final guessing; and implement multiple methods for each (summarized in Table 1). These methods are combined into a symbolic agent integrating possible-worlds belief tracking, information-gain-driven question selection, and MAP guessing; and an LLM agent that performs all three sub-tasks through structured prompting. Second, we build an ablation Prince agent that fixes belief tracking to possible-worlds belief tracking and final guessing to MAP estimation, while leaving question selection configurable between information gain, random, and LLM-based selection (with optional context of question-and-answer history and symbolic belief state). Third, we report three preliminary experiments: an LLM configuration sweep, a cross-agent comparison covering all combinations of Prince and non-Prince agents, and a component ablation that tests the contribution of belief tracking and that of question selection.

These experiments reveal several findings that motivate future work. First, the symbolic Prince’s accuracy drops sharply against strategic symbolic non-Prince agents. We attribute this to an opponent model mismatch between its uniform-answer likelihood model and the utility-maximizing behaviours of such non-Prince agents, which can be viewed as a mismatch in theory-of-mind (ToM) reasoning level between the Prince and non-Prince agents. Second, possible-worlds belief tracking provides consistent accuracy improvement regardless of the question-selection strategy. This suggests that it has independent value as a component and motivates retaining it in future agent design. Third, LLM-based question selection outperforms the information-gain heuristic when given access to the question-and-answer history, the symbolic belief state, or both, but underperforms even random question selection without such context. This indicates that the LLM’s value lies in interpreting structured information rather than in reasoning from scratch, and motivates a future neuro-symbolic design that assigns each sub-task to the component best suited for it.

Background

Game Tofu Kingdom

Tofu Kingdom is a social deduction game for 3–8 players (EverythingBoardGames 2018; Ultraboardgames 2026). One player is the Prince, who must identify the hidden Princess without knowing any role; all other players see the full role assignment. In games with fewer than 8 players, one role is placed as a center card with no player controlling it. The core mechanism is information asymmetry: the Prince reasons under uncertainty, while everyone else has full information but constrained communication.

Players belong to three factions: the Princess faction (must tell the truth; wins if the Prince finds the Princess), the Queen faction (must lie; wins if the Prince picks the Queen), and the Other faction (may answer freely; wins if the Prince picks neither). The game proceeds through three phases: Questioning, where each non-Prince player receives and an-

swers exactly one of three fixed question types (“Where is the Princess?”, “Who are you?”, or “Who is Player X?”); a Final Question to any player; and a final Guess.

We choose Tofu Kingdom as our testbed because it focuses purely on information gathering: as long as the Prince correctly identifies the roles of the other players, it can win. In addition, unlike Avalon or Werewolf, Tofu Kingdom has no interleaved voting or free-form discussion. Both questions and answers come from fixed, enumerable sets, which enables direct comparison between symbolic and LLM-based agents within a constrained action space.

Related Work

This work draws on two areas: AI agents for social deduction games with and without LLMs. Research ranges from pure LLM approaches, through purely formal methods, to hybrid systems; we review them in order.

Pure LLM approaches. AvalonBench (Light et al. 2023) benchmarks LLMs on social deduction. A ReAct-style GPT-3.5 agent playing as the Good team in Avalon won only 22% of games, worse than a simple rule-based bot at 38%. This suggests that prompting alone is insufficient for tasks that require strategic deception and social inference.

ReCon (Wang et al. 2023) improves on this through structured prompting, running two “contemplation” stages each turn: inferring others’ roles from game history, then reasoning about how they perceive the agent. This aids perspective taking, but as in all pure-LLM methods, the reasoning stays entirely inside the language model, with no formal structure to mitigate inconsistent or hallucinated inferences.

Formal and symbolic approaches without LLM. In the same area, there are several systems that perform social deduction using formal logic or game theory methods, without a large language model component.

Eger and Martens (Eger and Martens 2017, 2018, 2019) built Ostari, a general reasoning system based on Dynamic Epistemic Logic (DEL): a logical framework that models how agents’ beliefs change after epistemic events such as announcements or observations (van Ditmarsch, van der Hoek, and Kooi 2007). They apply Ostari to One Night Ultimate Werewolf (and other games such as Hanabi) in their researches. The game state is represented as a weighted set of possible worlds (Kripke 1963), where each world is a complete role assignment that the agent considers consistent with its observations. Physical actions (peek at a card, swap roles) are modelled as epistemic actions that update which worlds each agent considers accessible. Communicative actions (make a claim) do not add or remove possible worlds, but reduce weight on worlds inconsistent with the claim, which represents the possibility that the speaker lies. The central question of their work is commitment: once an agent has announced a claim (maybe a lie), how long should it keep that claim when new information is observed? They compare three strategies: always replan, replan only if the new plan is better, and never replan. The 2019 paper also evaluates strategies against human players. Ostari assigns heuristic weights to possible worlds based on how many observed statements each world is consistent with, but it selects

actions to approach fixed goals rather than to actively reduce uncertainty.

DeepRole (Serrino et al. 2019) is a neuro-symbolic method that uses a game-theoretic approach. It combines the Counterfactual Regret Minimization (CFR) algorithm with deep value networks and deductive reasoning over joint role assignments to play Avalon, and beats human players across thousands of games in The Resistance: Avalon. But DeepRole communicates only through structured votes, never in natural language, which means it cannot participate in the free-form discussion in real gameplay.

These formal approaches provide logical support or game theory guarantees, but none of them handle natural language interaction. This limits their applicability to richer social deduction game settings.

Hybrid approaches combining LLMs with structured reasoning. Some recent work combines LLMs for natural language processing with rule-based modules for belief inference or strategy computation.

GRAIL (Rahimirad et al. 2025) is the current state of the art for social deduction in the game The Resistance: Avalon. Rather than relying solely on an LLM for reasoning, this framework externalizes belief tracking to a Factor Graph, which maintains a probability distribution over each player’s role throughout the game. At each round, the LLM first extracts the hidden information from game dialogue and adjusts the prior probability over role assignment accordingly. The Factor Graph then uses max-product belief propagation to infer possible hidden role assignments with the adjusted prior probability by updating belief for each player’s role. A neural network is also trained on over 100,000 real human gameplay logs to estimate the conditional probabilities in the factor functions during Belief Propagation. The LLM is responsible for generating dialogue from belief to interact with other agents. In their experiments, the GRAIL framework outperforms the LLM agent and achieves a 67% win rate against human players in Avalon, while it uses far fewer tokens than pure LLM agents. Nevertheless, GRAIL has several limitations relevant to our work. First, since belief propagation on a loopy graph provides no formal guarantee of convergence to the correct solution. Second, the authors report that GRAIL is restricted to the Good team role: the Evil agents remain pure LLM-based and cannot reliably plan deception without a symbolic reasoning module. Third, the system does not reason about which questions or proposals would be most informative: it reacts to information as it arrives rather than actively seeking it (the action is only determined by LLM with heuristic instruction based on belief).

CSP4SDG (Xu et al. 2025) is the closest existing framework to our target task. It models role identification as a constraint satisfaction problem (CSP). This framework uses an LLM to extract four types of constraints from game logs: evidence (hard facts about role assignment), phenomena (fact observations), assertions (player claims), and hypotheses (inferred possibilities). The symbolic module uses hard constraints (evidence and phenomena) to prune impossible role assignments and uses soft constraints to assign weights

to the rest by a heuristic score combined with IG: the expected reduction in entropy (Quinlan 1986; Shannon 1948). Then the framework outputs both a calibrated posterior distribution over roles and a MAP estimate: the hypothesis with the highest posterior probability (Bishop 2006), which in CSP4SDG is the role assignment that maximizes the posterior. The posterior result is either used individually or supplied as a context to an LLM to support LLM generation. In the experiment, CSP4SDG outperforms pure LLM baselines on Avalon and Mafia game datasets. However, it is a passive inference module: it analyses observed evidence but does not actively choose what evidence to gather. In other words, this framework is designed to identify role assignments by analyzing the previously recorded game log rather than instructing automatic agents in real games.

DEL-ToM (Wu et al. 2025) combines DEL with LLMs to perform Theory of Mind (ToM) reasoning: the ability to reason about the mental states of others: their beliefs, intentions, and aims (Premack and Woodruff 1978). DEL-ToM decomposes belief inference into a sequence of update steps formalized in DEL, then it trains a PBM (a lightweight LLM in their experiments) verifier by pairing reasoning traces generated by the LLM with noise-free, step-level labels from a DEL simulator, which is automatic supervision without human annotation. At inference time, the LLM generates multiple candidate reasoning chains and uses the PBM to score each step in these chains. This framework can substantially improve the performance of small LLMs in theory of mind reasoning and also allows small models to match or exceed much larger ones. The idea of using possible world structure to track agents’ beliefs for theory of mind reasoning is related to non-Prince agents in our research. However, DEL-ToM focuses on passive belief inference in simple, non-deceptive narratives. It does not handle adversarial agents who deliberately lie, and it does not support action selection or planning.

Across all three categories, a consistent pattern emerges. Every approach trades off one desirable property for another. Pure LLM methods have flexibility but no logical guarantees. Pure formal methods have rigorous logical reasoning but no language capability. Hybrid systems combine both to some degree. However, existing approaches treat belief inference primarily as a reactive process: they update beliefs when evidence is observed, but do not use the maintained belief state to actively plan which information to gather.

Methodology

We decompose the agent’s decision process into three modules: belief tracking, question selection, and final guess selection. We also include a separate module for non-Prince answer selection. Although question selection and final guessing could be combined into a single “action selection” module, we treat them separately so that their individual contributions can be analyzed. Each module has multiple implementations, and different agent types correspond to different combinations of these modules (summarized in Table 1). We also implement a random agent as a baseline, which selects uniformly at random from the valid actions in every module except belief tracking.

The following notation is used throughout this section:

- w^* denotes the true world (known by all agents except the Prince)
- w_i is the i -th possible-world (a hypothesis the Prince holds)
- O represents the observation history
- $P(w_i | O)$ is the Prince’s belief on $w^* = w_i$, given observation history O
- Q represents a complete question (question type with target player and any additional parameter)
- a means the observed answer (or expected answer during simulations)
- $H(X)$ represents Shannon entropy

Belief Tracking Modules

The belief tracking module maintains the Prince’s belief over the hidden role assignment and updates it when a new answer is observed. There are two different implementations for this module.

No Explicit Belief Tracking In the pure LLM agent, belief tracking is performed implicitly: the LLM receives the question-and-answer history and an iterative reasoning summary carried from previous turns, and performs all inference internally through Chain-of-Thought (CoT) reasoning. Although this allows flexible reasoning, it provides no formal guarantee of logical consistency. The pure random agent performs no belief tracking at all.

Possible-world Belief Tracking

Following the possible-worlds semantics of Kripke (Kripke 1963), we model the epistemic state of Prince as a probabilistic Kripke frame: a weighted set of possible-worlds, where each possible-world is a complete assignment of roles to all players (and, when applicable, to the center card), and the weights represent the Prince’s belief distribution over these worlds. From each world, we derive each player’s faction and answer constraint, as well as the unique truthful answer to each question. This design is based on several ideas from the literature: using Possible-worlds semantics for belief tracking in theory of mind reasoning (Wu et al. 2025), identifying roles with constraint-based methods (Xu et al. 2025), and hybrid architectures that externalize belief inference to a probabilistic model (Rahimirad et al. 2025).

Possible-world Enumeration. The possible-world belief tracking module enumerates all legal permutations and assigns each an equal prior weight. A possible world is one complete assignment of the role cards to positions, with one role per non-Prince player and, when enabled, one role for the center card. Therefore, the number of worlds is $|W| = n!$, where n is the number of non-Prince roles. This full enumeration is feasible because Tofu Kingdom’s state space is small, with at most $7! = 5040$ worlds for 8 players and only 24 worlds in the simplified 5-player setting. The module can therefore enumerate the state space exhaustively without sampling or approximation.

Bayesian Belief Update. When the Prince observes answer a to question Q , the weight of each possible-world is updated using Bayes’ rule:

$$P(w_i | a, Q, O) \propto P(a | w_i, Q) \times P(w_i | O) \quad (1)$$

Worlds with zero likelihood are removed from the accessible set and the remaining weights are normalized.

Answer Likelihood Model (opponent model). The likelihood $P(a | w_i, Q)$ is determined by the faction answer constraint of the answering player in world w_i :

- For princess faction, the player must give the truthful answer. The likelihood is 1 if answer a matches the truth in w_i , and 0 otherwise.
- For Queen faction, the player must give an answer which differs from the truth. In the basic symbolic agent, we assume a uniform distribution over all valid lies (i.e. the agent assumes that all opponents choose answers randomly), the likelihood is:

$$P(a | w_i, Q) = 1/(K - 1) \quad (2)$$

where K is the number of valid answers, and the truthful answer is excluded.

- For Other faction, the player may freely answer. We assign a uniform distribution over all possible answers:

$$P(a | w_i, Q) = 1/K \quad (3)$$

This corresponds to a 0-level ToM opponent model (de Weerd, Verbrugge, and Verheij 2013): the module does not reason about how rational opponents might choose actions strategically; it only assumes random behaviour of opponents within each faction’s constraint. Theory of Mind is hierarchical: a level d agent models opponents as reasoning at level $d-1$, recursively. Modelling higher-level ToM opponents is left to future work.

Question Selection

The question selection module determines which question the Prince asks during the Questioning and Final Question phases.

Information Gain Maximization. This method selects questions by maximizing expected information gain about Princess Tofu’s identity:

$$\begin{aligned} IG(Q) = & H_{\text{prior}}(\text{Princess}) \\ & - \sum_{a \in A} P(a | Q, O) H_{\text{posterior}}(\text{Princess} | a, Q, O) \end{aligned} \quad (4)$$

where $H(\text{Princess})$ is the Shannon entropy of the distribution over which player (or center card) holds the Princess Tofu role card, and

$$P(a | Q, O) = \sum_i P(w_i | O) \times P(a | w_i, Q) \quad (5)$$

is the marginal probability that answer a appears. The method outputs the question with the highest information gain. This builds on the passive role identification in CSP4SDG (Xu et al. 2025), which uses information gain as part of the weighting for observed constraints; we extend it to an active-planning setting where the agent chooses which constraint (i.e. answer) to acquire.

LLM-Based Selection. This method queries an LLM with a prompt containing game rules, the current observation (question-and-answer history, the roles in the current game, and the current phase), an iterative reasoning summary from previous turns, and the list of valid actions. The LLM is instructed to reason step by step (CoT) before selecting an action, and returns a structured response consisting of a reasoning summary and the index of the chosen action. The reasoning summary is carried forward as cumulative context for subsequent turns. In the ablation configuration, this method can additionally receive belief state information from the possible-world belief tracking module (like role probability distributions or top possible-worlds) as part of the prompt to support question selection. The belief tracking module remains responsible for updating beliefs from observed answers; this method only uses belief state to determine question selection.

Random Selection. This method chooses a question randomly at uniform from the valid actions. In the ablation configuration, it can be combined with the possible-world belief tracking module so that beliefs are still updated even when the questions are selected randomly.

Final Guess

The final guess module determines the Prince’s identification decision in the Guessing phase.

LLM-Based Guess. This method outputs the guess using the same LLM prompting mechanism as in the “Question Selection” section.

MAP Estimation. This method outputs the candidate with the highest posterior probability of being Princess Tofu (i.e. the MAP estimate). This requires the possible-world belief tracking module to be active, and guarantees that the final guess is the Bayesian-optimal choice given the maintained belief state.

Random Guess. This method picks a candidate uniformly at random from the valid targets.

Non-Prince Answer Selection

The non-Prince answer selection module determines how a non-Prince agent responds when questioned by the Prince.

LLM-Based Answer. This method queries an LLM with the same prompting mechanism as in the “Question Selection” section, but receives the full role assignment (including the center card), the current question, and a faction-specific strategy hint.

Symbolic ToM Answer (Level-1 ToM). Inspired by the use of formal epistemic models to represent other agents’ belief in theory of mind reasoning (Wu et al. 2025), this method applies level-1 ToM: it simulates the Prince’s belief update under each candidate answer a and chooses the answer that maximizes a faction-specific value function $V(a)$. Let $P(w_i | a, Q, O)$ be the simulated Prince’s posterior belief if the agent were to reply with a . The method selects:

$$a^* = \arg \max_{a \in A_{\text{valid}}} V(a) \quad (6)$$

where A_{valid} is the set of answers allowed by the answering agent’s faction constraint. The value function V is defined as follows:

- For princess faction, the answering agent must only give the truthful answer, so no value function is required.
- For Queen faction, the method selects the answer that maximizes the posterior probability that the Princess role is assigned to the actual Queen Tofu in the Prince’s belief: that is, the lie most likely to cause the Prince to guess Queen Tofu mistakenly. Let j represents the player ID of the actual Queen Tofu in the true world. Then value function is:

$$V_{\text{Queen}}(a) = P(\text{Princess} = j | a, Q, O) \\ = \sum_{i: \text{all } w_i \text{ for Princess role at } j} P(w_i | a, Q, O) \quad (7)$$

- For Other faction, the method aims to make the Prince choose neither Princess Tofu nor Queen Tofu. Let j_P and j_Q be the player IDs of the actual Princess Tofu and Queen Tofu. It minimizes the combined posterior probability that the Prince guesses either of these targets. The method also uses a secondary strategy that aims to maximize the Prince’s remaining uncertainty about the Princess’s identity through maximizing normalized posterior entropy:

$$V_{\text{Other}}(a) = \lambda \left[1 - P(\text{Princess} = j_P | a, Q, O) \right. \\ \left. - P(\text{Princess} = j_Q | a, Q, O) \right] \\ + (1 - \lambda) \frac{H(\text{Princess} | a)}{\log_2 N} \quad (8)$$

where $\lambda \in [0, 1]$ is a tunable weight that balances the two strategies and N is the number of candidate Princess locations (players or center card).

Random Answer. This method picks an answer uniformly at random from those allowed by the answering agent’s faction constraint.

Agent Configurations

Different agent types correspond to different combinations of the modules. Table 1 summarizes the configurations of the four agent types implemented for our experiments.

Experiments

To identify the limitations of the basic method and compare it with the LLM-based agent, we conduct several experiments for analysis.

All LLM agents in our experiments use Qwen 3.5 (9B parameters, 8-bit quantized) (Qwen Team 2026) deployed locally via Ollama (Ollama n.d.) on GPU cloud instance (GPU L40s-16c (24GB vRAM) 16 vCPU 118GB). For the Basic symbolic Other-faction agent, the trade-off weight λ in Equation 8 is set to 0.5 in all experiments unless otherwise stated. We currently report two primary metrics: Prince accuracy (the proportion of games where the Prince correctly

Table 1: Agent configurations used in the experiments. symb. = symbolic; Ans. = Answer.

Agent Type	Belief Track.	Ques. Sel.	Final Guess	Non-Prince Ans.
Random	None	Random	Random	Random
LLM	None (internal)	LLM	LLM	LLM
Basic symb.	Possible-world	IG	MAP	Symb. ToM
Ablation (Prince only)	Possible-world	IG / Random / LLM	MAP	-

identifies Princess Tofu, i.e. the Princess faction win rate) and mean game duration in seconds. Accuracy measures reasoning quality while duration measures computational cost. Figure 1 and Tables 2 and 3 summarize the main quantitative results discussed in this section.

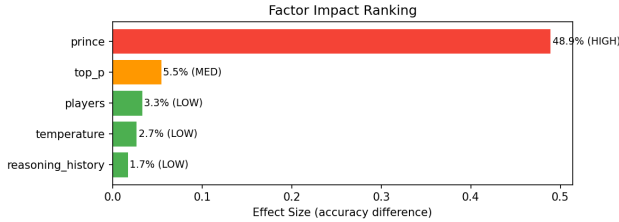


Figure 1: Effect sizes of different factors in Experiment 1.

Experiment 1: LLM Configuration Before comparing agent architectures, we first identified a robust LLM configuration as a baseline. We ran a factorial sweep over five factors: Prince agent type, non-Prince agent type, temperature (0.5, 0.7), top_p (0.8, 0.9), and reasoning history (enabled, disabled). This produced 32 combinations, with 20 games per configuration (640 games in total). The experiment used a simplified 5-player setting with no center card and only Princess-faction and Queen-faction roles.

As shown in Figure 1, the effect of Prince’s agent type dominated all other factors: LLM Prince averaged 75.8% accuracy while Random Prince averaged 26.9%, the effect size was 48.9 percentage points. All LLM parameters had effect sizes below 6%, with top_p showing the largest effect (5.5%). Temperature and reasoning history both contributed less than 3%, which suggests that the LLM’s game performance in accuracy is relatively robust to these settings within the tested ranges.

We validated the top 3 LLM configurations at full scale (7 players with center card, 100 games each). The top configuration (LLM vs LLM, reasoning history disabled, temp=0.5, top_p=0.8) achieved 65% accuracy but averaged 1253s per game. The second-ranked configuration (LLM vs LLM, reasoning history enabled, temp=0.5, top_p=0.9) achieved a comparable 63% but only 466s duration per game: it is more

Table 2: Cross-agent comparison results. Symb. = Symbolic; Rand. = random; Acc. = Accuracy; Opp. = Opponent.

Prince	Opp.	Acc.	Queen wins	Other wins	Mean time
Symb.	Rand.	95%	1%	4%	8s
LLM	Rand.	72%	4%	24%	696s
Symb.	LLM	66%	29%	5%	429s
LLM	LLM	63%	22%	15%	465.5s
LLM	Symb.	48%	22%	30%	630s
Symb.	Symb.	30%	52%	18%	10s

Note: the LLM Prince vs Random uses a different LLM setting (no reasoning history, temp=0.7, top_p=0.9) since it was run before the LLM baseline was finalized. Thus, its accuracy may not be directly comparable with other LLM Prince. Additionally, game durations with LLM are highly variable (individual games range from around 100s to over 3000s), so mean times should be interpreted as rough indicators rather than precise estimates.

than twice as fast for only a 2-percentage-point accuracy trade-off. Thus, we adopt this faster configuration as the LLM Prince baseline for subsequent and future experiments. Both configurations significantly exceed the random baseline (14% for the same game configuration with a different seed) but this still leaves considerable room for improvement.

Experiment 2: Cross Agent Comparison With the LLM baseline, we next compare it against the basic symbolic agent described in the “Methodology” section. We ran all Prince and non-Prince type combinations on the 7-player setting with center card (100 games each). The LLM Prince uses the baseline configuration selected in Experiment 1 for all conditions; the LLM non-Prince agents use the same model and settings. The detailed results are shown in Table 2.

Several patterns emerge from these results. First, the Basic symbolic Prince achieves nearly perfect accuracy against Random non-Prince opponents (95%) and also outperforms the LLM Prince (72%). When playing against LLM opponents, the Basic symbolic Prince is also better than LLM Prince (66% vs 63%). This indicates that Bayesian belief tracking combined with MAP guessing is effective when opponents do not employ targeted strategic deception.

Second, Symbolic against Symbolic accuracy drops to 30%, barely above the random baseline in the full scale (14%). We attribute this to a ToM-level mismatch between the Prince and the non-Prince agents. The Basic symbolic Prince’s belief update uses a 0-level ToM opponent model (Equations 2–3): it assumes Queen-faction players choose lies randomly and Other-faction players choose randomly over all valid answers. However, the Basic symbolic non-Prince agents apply level-1 ToM (Equations 7–8): the Queen agent selects the lie that maximizes the probability of the Prince guessing Queen Tofu, and the Other agent selects the answer that both minimizes the probability of the Prince

guessing Princess or Queen and maximizes the Prince’s remaining uncertainty over the Princess’s identity. This mismatch causes the Prince to systematically underweight answers that a rational level-1 opponent would choose.

Third, the two agent types have different strengths depending on their role. As non-Prince opponents, Basic symbolic agents are substantially more challenging than LLM agents when playing against both Prince types. Against LLM non-Prince opponents, the LLM Prince achieves 63% accuracy and the Basic symbolic Prince achieves 66% accuracy; while replacing those opponents with Basic symbolic agents, their accuracy drops to 48% and 30% respectively. This may indicate that, in the structured setting, the symbolic agent’s explicit payoff-maximizing strategy leads to more effective deception than the LLM, though this difference may also be caused by other factors such as the LLM prompt design or behavioural patterns of the LLM that the Prince agent may exploit. However, as Prince, the pattern partially reverses: against Basic symbolic opponents, the LLM Prince (48%) performs much better than the Basic symbolic Prince (30%). This is likely because the LLM is not rigidly bound to the 0-level ToM assumption (i.e. the mismatched likelihood model) and can partially adapt to unexpected answer patterns, even without an explicit opponent model. In contrast, the Basic symbolic Prince relies entirely on its mismatched likelihood model and cannot recover from this systematic error.

Fourth, there is a large efficiency gap between symbolic and neural approaches: pure Symbolic games complete in around 10 seconds, while configurations involving LLM agents range from several minutes to over an hour. However, LLM inference time is highly variable across games, so the mean durations in the table should be treated as rough estimates rather than reliable rankings between LLM involved configurations. The overall pattern is clear: symbolic computation is substantially faster, but more precise comparisons between different LLM configurations would require larger sample sizes to be meaningful.

Experiment 3: Component Ablation The cross-agent comparison shows that both LLM and Symbolic approaches have distinct strengths and weaknesses. To understand which components drive performance, we conducted a controlled ablation experiment. The key idea is that the possible-world belief tracking and MAP guess are always active; only the question-selection method varies. This isolates the value of question selection from the value of structured belief tracking.

We tested 12 configurations (3 question strategies, 2 history settings, 2 belief-state settings) for 100 games each (7 players, center card, 1200 games in total). To make it comparable, all opponents are fixed Basic symbolic non-Prince agents. Note that the possible-world belief tracking and MAP guess is functionally equivalent to the full Basic symbolic Prince in Table 2. The difference in accuracy (37% mean here and 30% in Table 2) may be attributed to different random seeds. A larger sample size would be required to determine whether the difference is statistically meaningful.

For IG and Random strategies, the context flags are ig-

Table 3: Component ablation results.

Question Selection	Context Given	Accuracy	Mean Time/Game
LLM	history + belief state	48%	214s
LLM	history only	46%	407s
LLM	belief state only	45%	129s
IG	–	37% (mean)	10s
Random	-	31% (mean)	0.3s
LLM	no context	25%	203s

nored; variation across their sub-conditions reflects seed variance only (32–41% for IG, 26–33% for Random).

Several findings can be derived from these results:

1. Possible-world belief tracking alone does contribute. Even with random questions, the Kripke-based MAP guess achieves 31%, well above the 14% random baseline. This confirms that Bayesian belief tracking benefits the agent regardless of the question-selection strategy.
2. IG-based question selection contributes moderately. IG improves the mean accuracy from 31% (random) to 37%, an improvement of approximately 6 percentage points. This gain is smaller than might be expected from a theoretically grounded approach. Two factors may contribute to this result: the IG computation relies on the 0-level ToM opponent model for belief updates, so its estimate of which question is most informative may be inaccurate against Basic symbolic opponents; and the current IG method uses a greedy single-step selection that does not take into account how the order of questions affects the result.
3. LLM question selection outperforms IG when given sufficient context. With both history and belief state provided, LLM-based selection achieves higher performance (48%) than the IG based selection (averaged in 37%). This means that the LLM may extract useful signals from the provided context that the IG heuristic does not capture. However, we have not determined the source of this advantage: possible explanations include implicit multi-step planning in LLM reasoning, avoiding the mismatched opponent model, or more diversity in question choice. Further analysis is required in our future work.
4. Context is critical for LLM performance. Without any context (no history, no belief state), the LLM scores only 25%, worse than random questions with possible-world belief tracking. Since both conditions use the same MAP guess, this comparison isolates question-selection quality: it shows that the LLM provides no advantage in question selection when situational information is removed. Its value lies in interpreting and integrating the provided context.

Future Work

The experiments above motivate four directions for future work, addressing both the symbolic baseline’s main weakness and the LLM-symbolic integration identified in the ablation study.

Higher-Level ToM Opponent Model. The symbolic Prince’s dominant failure mode is its low accuracy (30% and 37%) against symbolic non-Prince agents, which can be viewed as a level-0 ToM Prince facing level-1 ToM opponents. The next step is to replace the random answer-likelihood model with a level- d opponent model that recursively assumes opponents reason at level $d-1$, following the ToM hierarchy (de Weerd, Verbrugge, and Verheij 2013). At level 1, for example, the Prince assumes each Queen-faction opponent would choose the lie that maximizes the probability of the Prince guessing that opponent as Queen Tofu, which is the same value function used by our non-Prince symbolic agents. Concretely, suppose player j is a Queen-faction player and the Prince asks j Q2 before ruling out j as the Princess. In that situation, “I am Princess Tofu” is the only Q2 answer consistent with the hypothesis that j is the Princess, who must answer truthfully. Under V_{Queen} (Eq. 7), this answer can therefore be a value-maximizing lie for player j , and a level-1 Prince would treat it as a self-serving strategic answer rather than as one arbitrary lie among $K-1$ possible answers. Exact enumeration is tractable at level 1 but its cost grows by a factor of roughly $K|W|$ per additional level, where $|W|$ is the number of possible worlds and K is the average number of valid answers for a question. If runtime becomes impractical, the belief updates can be restricted to the top- k highest-probability worlds. We expect level-1 reasoning to substantially reduce the gap between Symbolic vs Symbolic (30%) and Symbolic vs Random (95%) accuracy.

However, whether deeper reasoning helps is less clear. Because our non-Prince agents are level-1, a level-2 Prince might over-attribute sophistication and re-introduce a mismatch in the opposite direction, suggesting that opponent depth may need to be matched rather than simply maximized, a question we leave open.

Multi-Step Question Planning. The current information-gain heuristic is greedy and does not account for how question order affects subsequent observations. A possible extension is expected-max search over question-answer trees, with opponent responses determined by the level- d ToM model. For a 7-player game with center card, a 2-step lookahead expands to approximately 10^5 nodes, which remains tractable with alpha-beta pruning. Deeper lookahead can be handled by MCTS style sampling, based on ideas from recent work on planning in social deduction games (Zhang et al. 2025).

Neuro-Symbolic Integration. The ablation results suggest that the LLM and symbolic module have complementary strengths: possible-world belief tracking with MAP guessing improves accuracy regardless of question strategy, while LLM-based question selection outperforms the IG method when given enough context. This motivates a more

extensive integration than the fixed pipeline of the current ablation agent. The integrated agent could work as follows: 1. the belief module supplies structured belief context (role distributions or top- k worlds) to the LLM for question selection 2. the LLM may query the belief module as a tool to simulate counterfactual updates (e.g. “if I ask Q and receive a, how does my belief change?”), which allows it to reason about question value without performing Bayesian computation itself; and 3. the belief module flags high-uncertainty situations to selectively invoke the LLM, this better controls inference cost.

Cross-Domain Validation. Beyond Tofu Kingdom, a longer-term goal is to port our framework to other games such as The Resistance: Avalon (Avalon-game.com 2026), where information gathering must be balanced against mission outcomes and where the game involves open-ended dialogue. The possible-world belief-tracking component transfers directly: Avalon’s role-assignment space is at most $C(10,4) = 210$ worlds, well within exact-enumeration range, but action selection must be extended to weigh information value against expected mission outcomes. Our framework can be compared against GRAIL (Rahimirad et al. 2025) and ReCon (Wang et al. 2023) as established Avalon baselines. A more extensive experiment is to transplant our IG-driven action-selection module into existing belief frameworks that are currently passive: like GRAIL (Rahimirad et al. 2025) and CSP4SDG (Xu et al. 2025), both of which react to evidence when it is observed rather than actively plan what evidence to gather, which can test the value of active information gathering as a general purpose component, independent of our specific implementation.

Conclusion

We presented a modular framework for active information gathering in Tofu Kingdom, decomposing the Prince into belief tracking, question selection, and final guessing, plus a module for non-Prince answer selection, and instantiating these as a symbolic agent and an LLM agent. A controlled ablation showed that possible-worlds belief tracking provides consistent improvement regardless of question strategy, while LLM-based question selection is competitive with the information-gain heuristic only when given enough context. Against strategic symbolic opponents, the symbolic Prince’s accuracy drops sharply, which we attribute to a mismatch between its level-0 opponent model and the level-1 strategies the non-Prince agents actually use. Future work can improve opponent modelling, add multi-step planning, transplanting the IG action-selection module into existing frameworks, and test the framework in broader settings such as Avalon.

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